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DISORDERS OF OESOPHAGUS IN CHILDREN

Esophageal Atresia

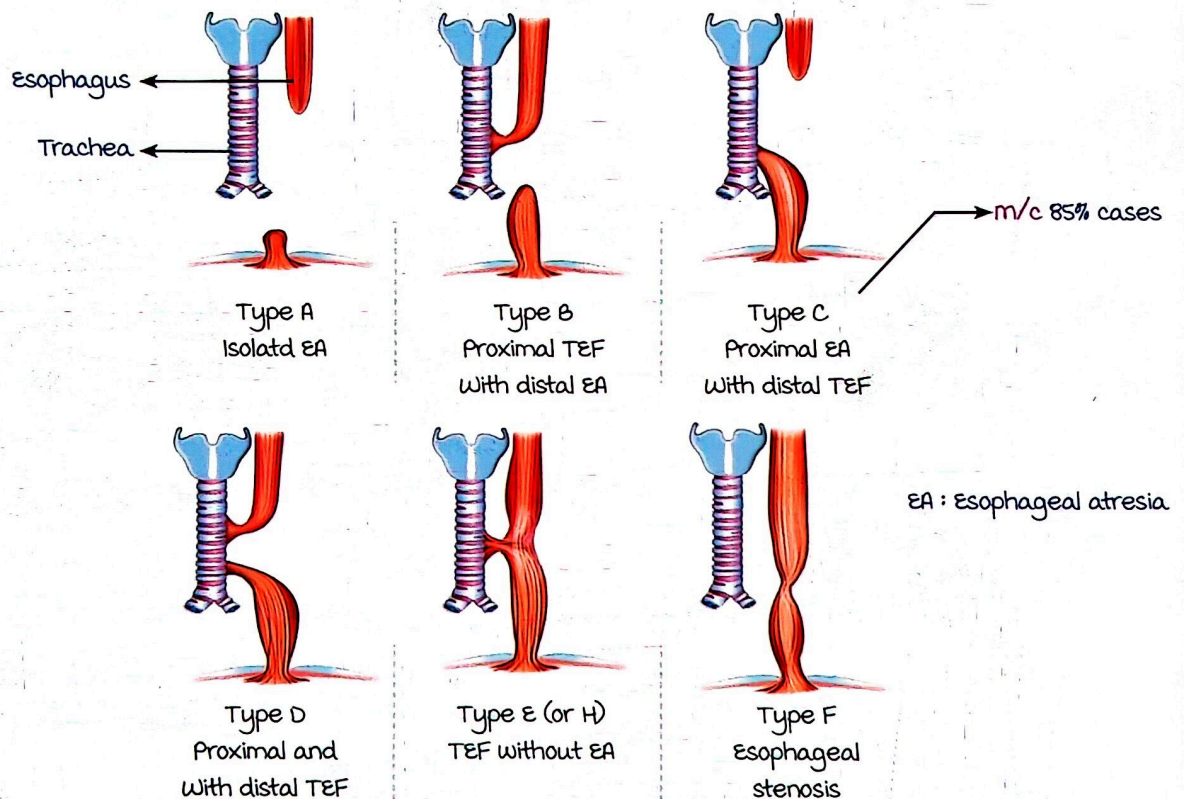
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Discontinuity in the esophagus.

ASSOCIATED ANOMALIES :

1. Tracheo-esophageal Fistula (TEF) : m/c defect in mesenchymal separation b/w trachea & esophagus.

- Gross classification :



a. VACTERL association :

- Vertebral anomalies.
- Imperforate Anus.
- Cardiac defects.
- Tracheo Esophageal fistula.
- Renal anomalies.
- Limb anomalies.

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Feedback

CLINICAL FEATURES :

- Vomiting $\xrightarrow{\text{multiple episodes}}$ Aspiration pneumonitis.
- Excessive frothy salivation.
- Antenatal : Polyhydramnios (Inability to swallow amniotic fluid).

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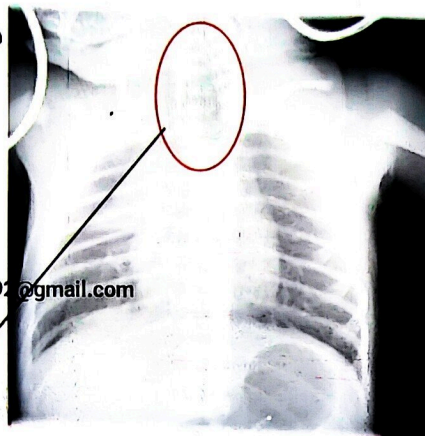
INVESTIGATIONS :

Chest X-ray :

- Resistance encountered during insertion of nasogastric (NG) tube.

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Coiled nasogastric tube in
proximal esophagus



Chest X-ray

TREATMENT :

Surgery : Anastomosis of esophageal ends + Division & ligation of TEF.

Gastro Esophageal Reflux Disorder (GERD)

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CLINICAL FEATURES :

- Age of onset : < 6 months (Infancy).
- Recurrent vomiting $\left\{ \begin{array}{l} \rightarrow \text{Aspiration} \\ \rightarrow \text{malnutrition} \\ \rightarrow \text{Failure to thrive} \end{array} \right.$ (D/t short esophagus)
- Sandifer syndrome : Characteristic.
 - Spasmodic dystonia (Posturing of the back muscles) : Opisthotonus.

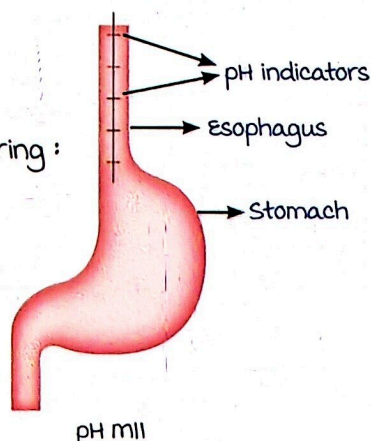
Note : Opisthotonus is also seen in $\left\{ \begin{array}{l} \text{Neonatal tetanus} \\ \text{Bilirubin-induced neurological damage (BIND)} \end{array} \right.$

INVESTIGATIONS :

1. 24-hr pH monitoring of esophagus :
 - pH ≤ 4 indicates GERD.
2. pH-mII (multichannel intraluminal impedance) monitoring :
 - Best test.
 - measures the level up to which reflux occurs.

Upper GI endoscopy :

Erosions & breaks in oesophageal mucosa seen.



pH mII

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TREATMENT :

1. Diet & lifestyle modifications : $\xrightarrow[\text{improvement}]{\text{No}}$ 2. Pharmacological : $\xrightarrow[\text{improvement}]{\text{No}}$ 3. Surgery :
- Small, frequent feeds.
 - Thickening of feeds.
 - Avoid caffeine, spicy foods
 - Avoid chocolates (In older children).
 - PPIs : Omeprazole, Lanzoprazole.
 - Duration of Rx : 3 months (To prevent recurrence).
 - Done for refractory GERD.
 - Nissen fundoplication.

Nissen fundoplication :

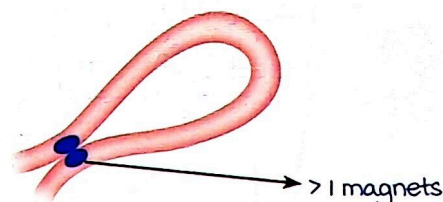
Part of fundus wrapped around lower part of esophagus $\rightarrow$ $\uparrow$ lower esophageal sphincter (LES) tone/pressure $\rightarrow$ $\downarrow$ reflux

Foreign Bodies in Esophagus

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FEATURES :

- Age : ≤ 3 years.
- Common foreign bodies : Coins, nuts, small toys.
- **Dangerous foreign bodies :**
 - Button/batteries $\rightarrow$ Coagulative necrosis of esophagus.
 - multiple magnets $\rightarrow$ Stick to each other across bowel loops $\rightarrow$ Obstruction $\rightarrow$ perforation.
 - Sharp objects.

**NATURAL COURSE :**

- most of foreign bodies ($> 95\%$) will pass through & get excreted.

Sites of impaction :

- Cricopharynx.
- mid-esophagus (At level of tracheal bifurcation).
- Right above LES.

CLINICAL FEATURES :

- H/o ingestion of foreign body may be absent.
- may be asymptomatic.
- If symptomatic :
 - Drooling of saliva.
 - Refusal of feeds.
 - Chest discomfort/pointing towards discomfort.
- Respiratory complaints : Recurrent cough/wheeze (D/t pressure effect on trachea).

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Disorders of Esophagus in Children

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INVESTIGATIONS :

1. X-ray : Only radio-opaque foreign bodies visualised.
2. Contrast study of esophagus/esophagogram : Filling defect seen.

MANAGEMENT :

1. Conservative : Blunt object & asymptomatic.
 - Observe for 24 hours.
2. Indications for removal by esophagoscopy :
 - Symptomatic.
 - Impacted for > 24 hrs.
 - Batteries/magnets/sharp objects.